S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Session-2025-26 (Even)

TAE-I Project Based Learning

Topic Name: Weather Prediction Using Bayesian Network

Submitted By

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AI Technique(s) (2)	Problem Representation (4)	Tools Explored (2)	Outcome (2)	Total (10)
Sign of Student:			Sign of course In-charge :	

1. INTRODUCTION

Weather prediction is one of the most important applications of Artificial Intelligence in real life. It helps in forecasting atmospheric conditions such as rainfall, temperature, and humidity. Traditional weather prediction techniques often rely on deterministic models, which may not effectively handle uncertainty present in real-world environments.

Artificial Intelligence provides a better approach by using probabilistic models that can deal with uncertainty and incomplete data. One such model is the Bayesian Network, which is a graphical representation of probabilistic relationships among variables. It uses Bayes' Theorem to update the probability of an event based on new evidence.

In this project, a Bayesian Network is used to predict whether it will rain or not. The model considers key weather parameters such as cloudiness and humidity and determines their influence on rainfall. The relationships between these variables are represented using a Directed Acyclic Graph (DAG), and Conditional Probability Tables (CPTs) are used to define these relationships.

2. PROBLEM STATEMENT

Weather forecasting involves predicting future atmospheric conditions, which is inherently uncertain due to the dynamic nature of environmental factors. Traditional models may not accurately capture this uncertainty.

The problem addressed in this project is to predict the likelihood of rainfall based on observed conditions such as cloudiness and humidity using a probabilistic approach. The model should provide probability-based output rather than fixed predictions

3. OBJECTIVES

- To understand Bayesian Networks
- To model weather prediction using probability
- To represent dependencies using a Directed Acyclic Graph (DAG)
- To implement the model using Python
- To perform inference and predict rainfall probability
- To visualize results using graphs and charts

4. FEASIBILITY STUDY

The proposed system is feasible as it uses a simple probabilistic model that requires minimal computational resources. It can handle uncertainty effectively and does not require large datasets. The implementation using Python and pgmpy library is efficient and easy to understand. The system can also be extended easily by adding more variables.

5. METHODOLOGY / AI TECHNIQUE

The project uses Bayesian Inference as the main Artificial Intelligence technique. A Bayesian Network is constructed where nodes represent variables such as Cloudy, Humidity, and Rain, and edges represent dependencies between them.

Conditional Probability Tables (CPTs) are defined to represent the relationships between variables. Using inference techniques like Variable Elimination, the probability of rain is calculated based on given conditions.

6. REPRESENTATION (DAG)

The Bayesian Network is represented using a Directed Acyclic Graph (DAG), where Cloudy and Humidity influence Rain. This means that the probability of rainfall depends on these two factors.

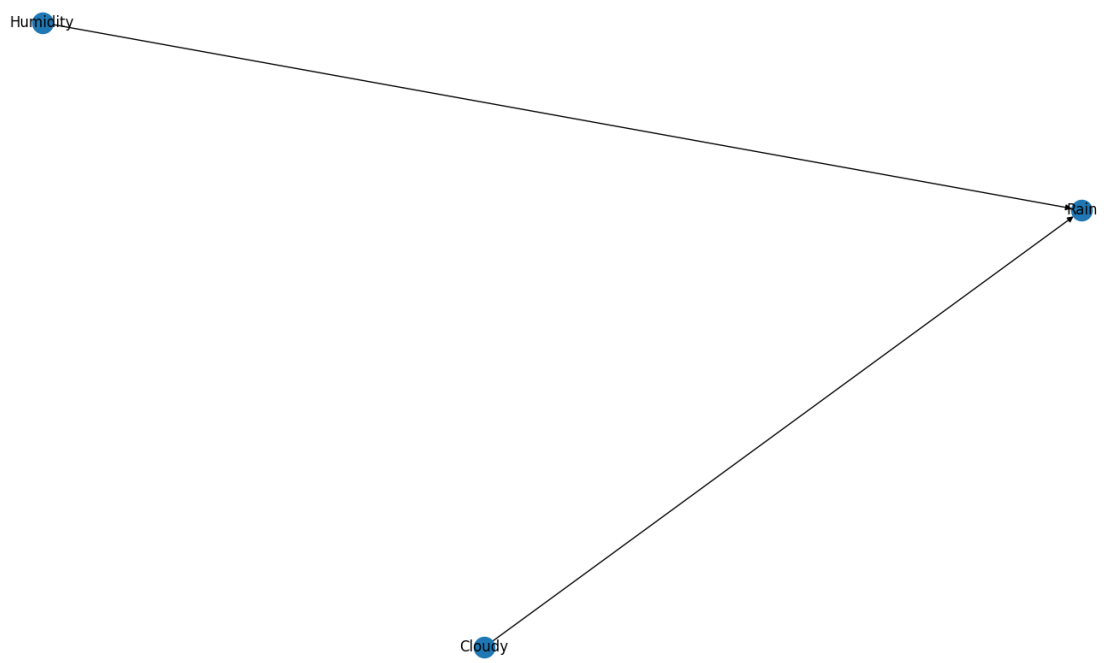
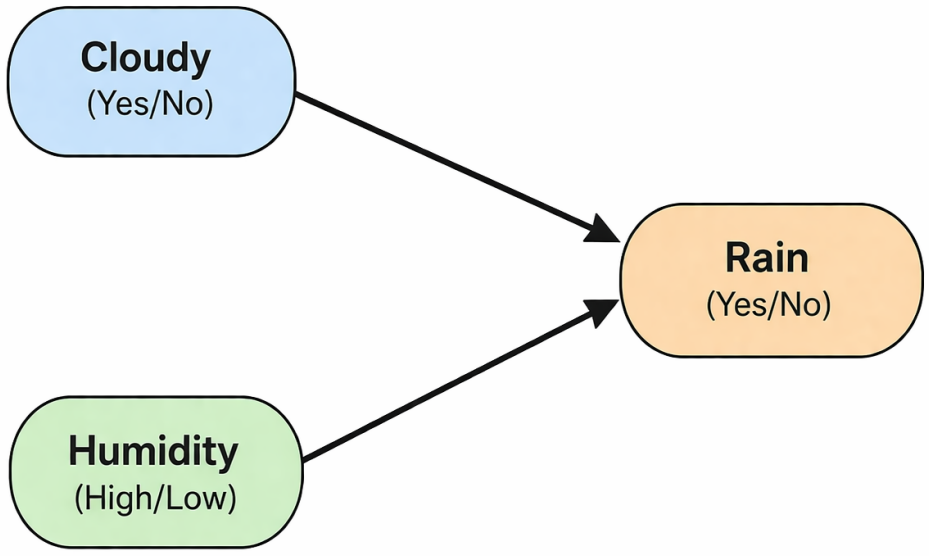


Figure 1: Bayesian Network DAG for Weather Prediction



DAG of Bayesian Network for Weather Prediction

7. CONDITIONAL PROBABILITY TABLES (CPT)

The following tables represent the probability distribution of each variable and their dependencies.

TABLE 1: Cloudy

Value	Probability
Yes	0.5
No	0.5

HUMIDITY TABLE

Value	Probability
High	0.6

Low	0.4

RAIN TABLE

Cloudy	Humidity	P(Rain = Yes)
Yes	High	0.9
Yes	Low	0.7
No	High	0.6
No	Low	0.1

8. IMPLEMENTATION

The model is implemented using Python programming language. The pgmpy library is used to create the Bayesian Network. The implementation includes defining the structure of the network, assigning probabilities using CPTs, and performing inference.

Code:

```
from pgmpy.models import DiscreteBayesianNetwork

from pgmpy.factors.discrete import TabularCPD

from pgmpy.inference import VariableElimination

import matplotlib.pyplot as plt

import networkx as nx
```

```
# Step 1: Define Bayesian Network structure
model = DiscreteBayesianNetwork([
    ('Cloudy', 'Rain'),
    ('Humidity', 'Rain')
])
# Step 2: Define Conditional Probability Tables (CPT)
# Cloudy
cpd_cloudy = TabularCPD(
    variable='Cloudy',
    variable_card=2,
    values=[[0.5], [0.5]]
)


# Humidity
cpd_humidity = TabularCPD(
    variable='Humidity',
    variable_card=2,
    values=[[0.6], [0.4]]
)
# Rain (depends on Cloudy and Humidity)
```

```
cpd_rain = TabularCPD(
    variable='Rain',
    variable_card=2,
    values=[
        [0.9, 0.7, 0.6, 0.1],   # P(Rain = No)
        [0.1, 0.3, 0.4, 0.9]    # P(Rain = Yes)
    ],
    evidence=['Cloudy', 'Humidity'],
    evidence_card=[2, 2]
)


# Step 3: Add CPDs to model
model.add_cpds(cpd_cloudy, cpd_humidity, cpd_rain)
# Step 4: Check model validity
print("Model Valid:", model.check_model())
# Step 5: Perform inference
infer = VariableElimination(model)
result = infer.query(variables=['Rain'], evidence={'Cloudy': 1})
print("\nProbability of Rain given Cloudy = Yes:")
print(result)
```

```
# Step 6: Draw Bayesian Network Graph

G = nx.DiGraph()

G.add_edges_from([('Cloudy', 'Rain'), ('Humidity', 'Rain')])

nx.draw(G, with_labels=True)

plt.title("Bayesian Network DAG")

plt.show()

# Step 7: Plot Bar Chart

labels = ['Rain', 'No Rain']

values = [0.6, 0.4]

plt.bar(labels, values)

plt.title("Rain Prediction Probability")

plt.show()
```

Search

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Users > sukriti > Desktop > AI_Project_Bayesian_Weather > code.py > ...

```python
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination
import matplotlib.pyplot as plt
import networkx as nx

# Step 1: Define structure (DAG)
model = DiscreteBayesianNetwork([
    ('Cloudy', 'Rain'),
    ('Humidity', 'Rain')
])

# Step 2: Define probabilities
cpd_cloudy = TabularCPD(variable='Cloudy', variable_card=2,
                        values=[[0.5], [0.5]])

cpd_humidity = TabularCPD(variable='Humidity', variable_card=2,
                          values=[[0.6], [0.4]])

```

PROBLEMS 5 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | Python

```
| Rain     |   phi(Rain) |
+==========+=============+
| Rain(0)  |      0.4000 |
+----------+-------------+
| Rain(1)  |      0.6000 |
+----------+-------------+
/usr/bin/python3 /Users/sukriti/Desktop/AI_Project_Bayesian_Weather/code.py
/usr/bin/python3 /Users/sukriti/Desktop/AI_Project_Bayesian_Weather/code.py
/usr/bin/python3 /Users/sukriti/Desktop/AI_Project_Bayesian_Weather/code.py
```

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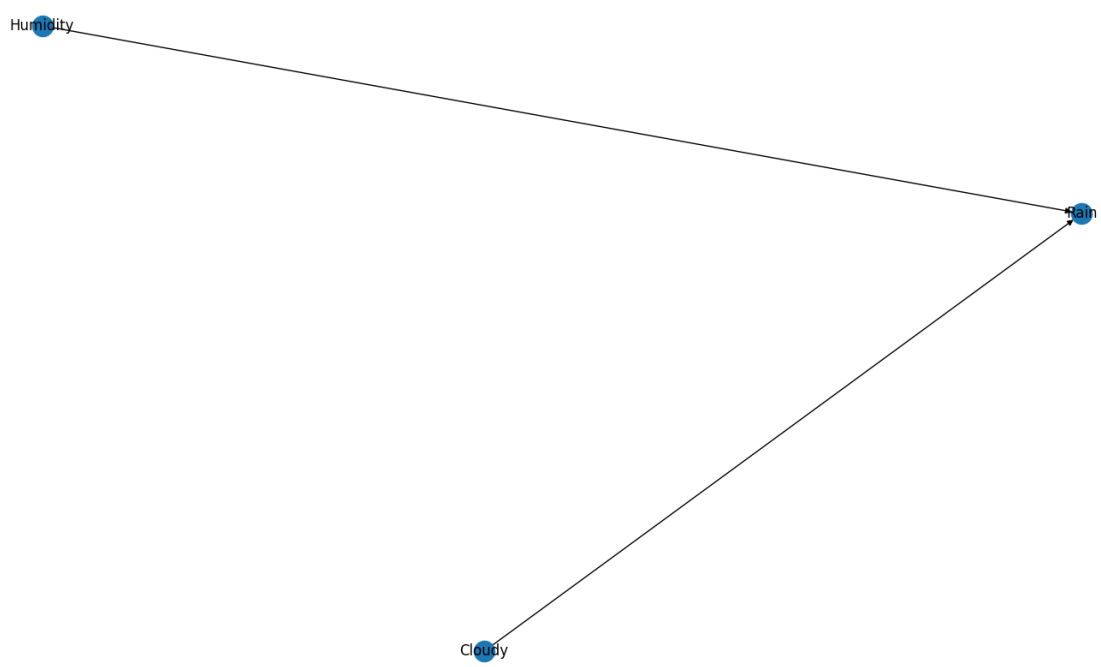

Figure 2: Python Implementation of Bayesian Network

9. OUTPUT

The Bayesian Network model successfully computes the probability of rainfall based on given conditions. In this project, when the condition "Cloudy = Yes" is provided, the model performs inference and calculates the probability distribution of the Rain variable

The output obtained from the system is:

Probability of No Rain = 0.4
Probability of Rain = 0.6

This result indicates that when the weather is cloudy, there is a higher likelihood of rainfall compared to no rainfall. The model uses probabilistic reasoning, which helps in handling uncertainty and provides more realistic predictions.

The output is also visualized using graphical representations. A Directed Acyclic Graph (DAG) is used to show the relationship between variables such as Cloudy, Humidity, and Rain. Additionally, a bar chart is used to represent the probability distribution of rainfall, making the results easier to understand.

10. CONCLUSION

The project successfully demonstrates the use of a Bayesian Network for weather prediction. The model uses probabilistic reasoning to determine the likelihood of rainfall based on input conditions such as cloudiness and humidity.

The results show that Bayesian inference is effective in handling uncertainty and provides realistic, probability-based predictions instead of fixed outcomes. The relationships between variables are clearly represented using a Directed Acyclic Graph (DAG), which helps in understanding how different factors influence rainfall.

The graphical visualization and probability charts further enhance the interpretation of results. Overall, this project highlights the importance of probabilistic models in Artificial Intelligence and their practical application in real-world problems such as weather forecasting.

12. REFERENCES

- [1] pgmpy Documentation, "pgmpy: Probabilistic Graphical Models using Python," Available: https://pgmpy.org
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